

Restaurant Sales Performance Analysis Using Excel and Power BI

Project Overview and Objective

Project Overview

This project focuses on cleaning, transforming, and analyzing restaurant sales data using Microsoft Excel and developing an interactive Power BI dashboard to identify sales trends, customer preferences, restaurant performance, and business insights for effective decision-making.

Project Objective

The primary objective of this project is to perform data preprocessing using Microsoft Excel, analyze restaurant sales performance using Power BI, and generate meaningful business insights through interactive visualizations.

Data Sources

- Source Description & Timeline: Restaurant Sales Dataset (2024–2025)
- Domain: Restaurant Sales Analytics / Food & Beverage Industry

Problem Statement

The objective of this project is to analyze restaurant sales data and generate meaningful business insights by answering the following business questions:

- What is the overall sales performance of the restaurant?
- Which branch, month, and order type generated the highest sales?
- Which menu items are the best-selling and least-selling products?
- Why are weekend sales higher than weekday sales?
- Why is the Lincoln Park branch performing below average?
- Why do some menu items and branches perform better than others?
- Which months and menu items are expected to generate higher sales in the future?

- How will customer demand and sales trends change in the coming months?
- What strategies should be implemented to improve low-performing branches, menu items, and customer satisfaction?
- How can inventory, staffing, and marketing be optimized to maximize sales and business performance?

Attribute (Column / Features) Details

Attribute Name	Data Type	Description
Transaction ID	Integer /Text	Unique ID for each customer Transaction.
Order Date	Date	Date when the order was placed
Restaurant Location	Text	Branch or Location where the order was placed.
Order Type	Text	Type of order (Dine-in, Takeaway, Delivery)
Menu Item	Text	Name of the food item ordered
Category	Text	Food Category (Main course, Beverage, Dessert, etc.)
Quantity	Integer	Number of Items ordered.
Unit Price	Decimal/Currency	Price of one unit of the menu item.
Order Total	Decimal/Currency	Total amount of the order(Quantity * Unit price)
Customer Rating	Decimal	Rating Given By the customer

Tools & Technologies

Microsoft Excel

- Data Cleaning • Data Transformation • Remove Duplicates • Sorting & Filtering • Conditional Formatting • Power Query • Pivot Tables

Power BI

- Data Modelling • DAX Calculations • Interactive Dashboard Development • KPI Cards • Charts and Visualizations • Slicers and Filters

Data Pre-Processing

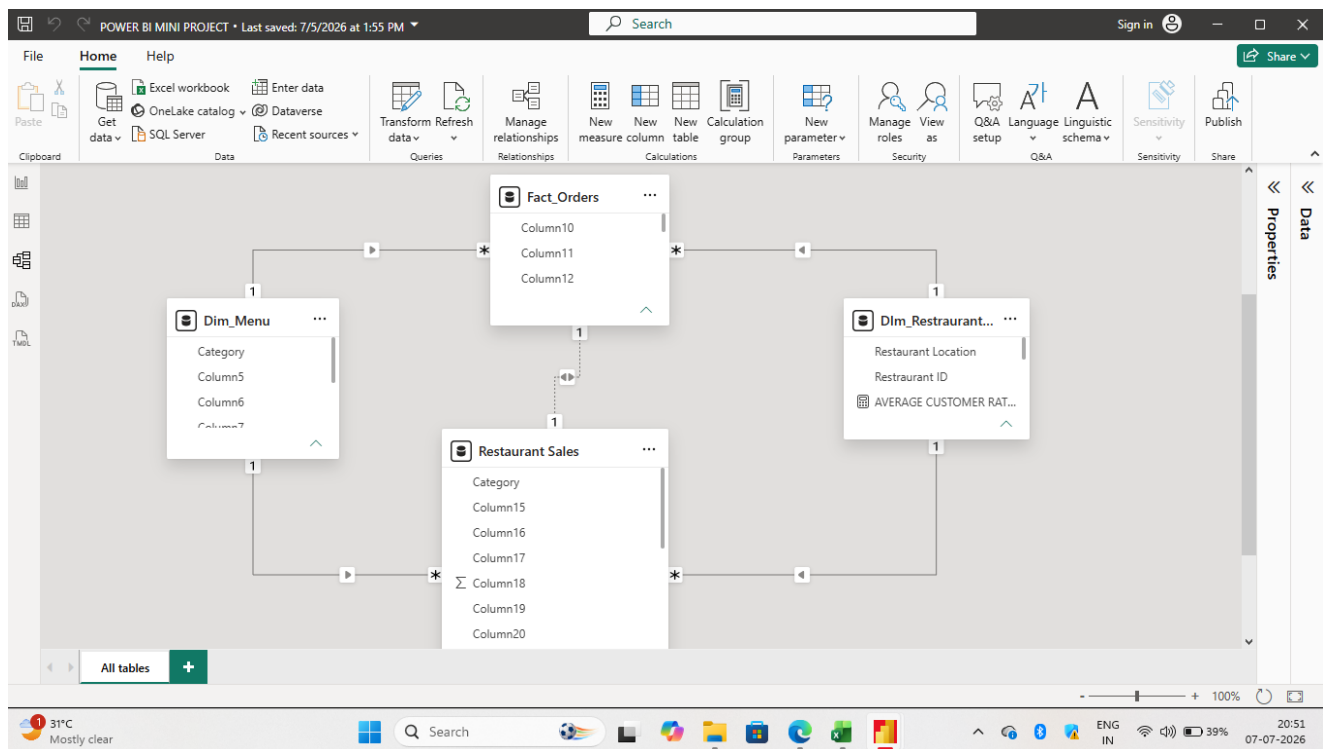
Tasks Performed

- Removed duplicate records.
- Handled missing values.
- Standardized data formats.
- Created calculated columns.
- Performed sorting and filtering.
- Generated Pivot Tables for data summarization.
- Created Fact and Dimension tables.
- Imported the cleaned dataset into Power BI.

Data Modelling and DAX (Power BI)

Data Model

- Established relationships between Fact and Dimension tables.
- Defined appropriate cardinality.
- Built an optimized data model for reporting.



Calculated Columns & DAX Measures

Total Sales =

SUM('Restaurant Sales'[Order Total])

Total Orders =

DISTINCTCOUNT('Restaurant Sales'[Order ID])

Total Quantity Sold =

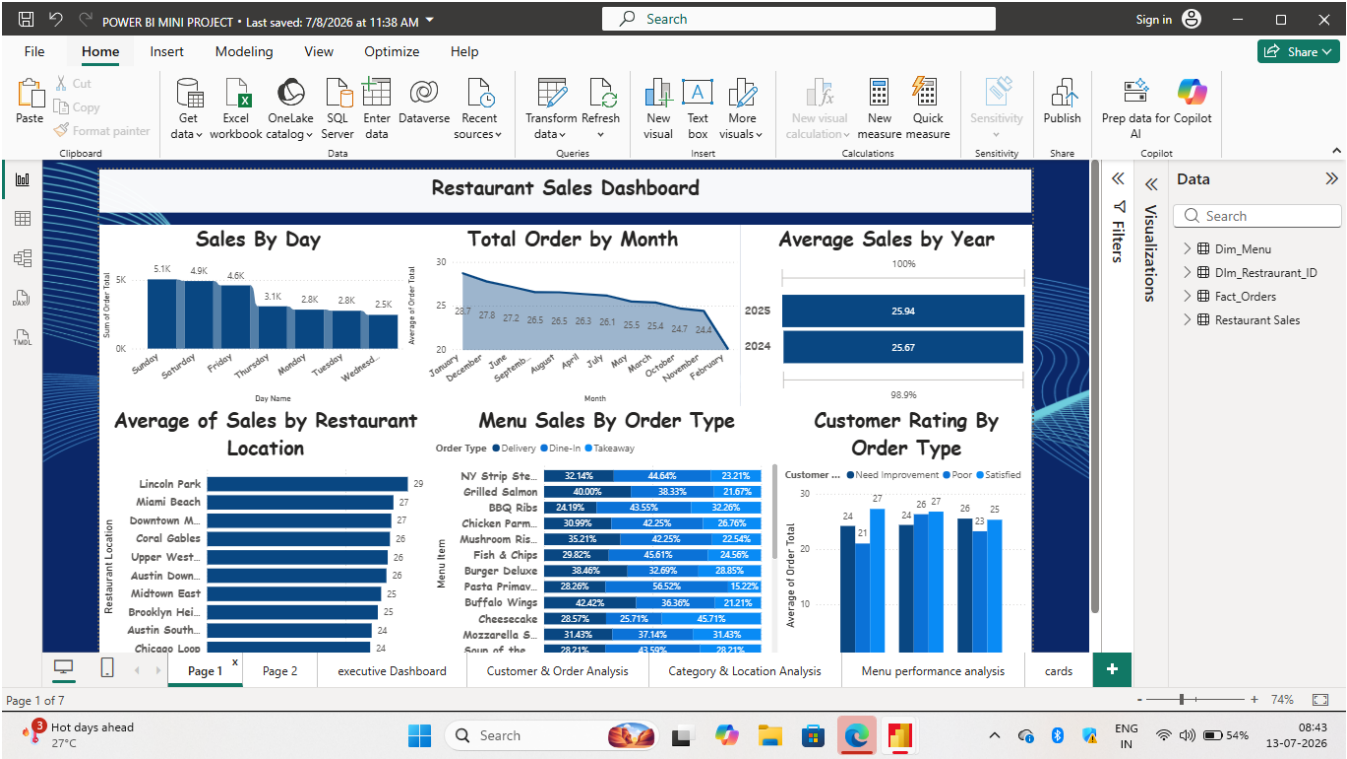
SUM('Restaurant Sales'[Quantity])

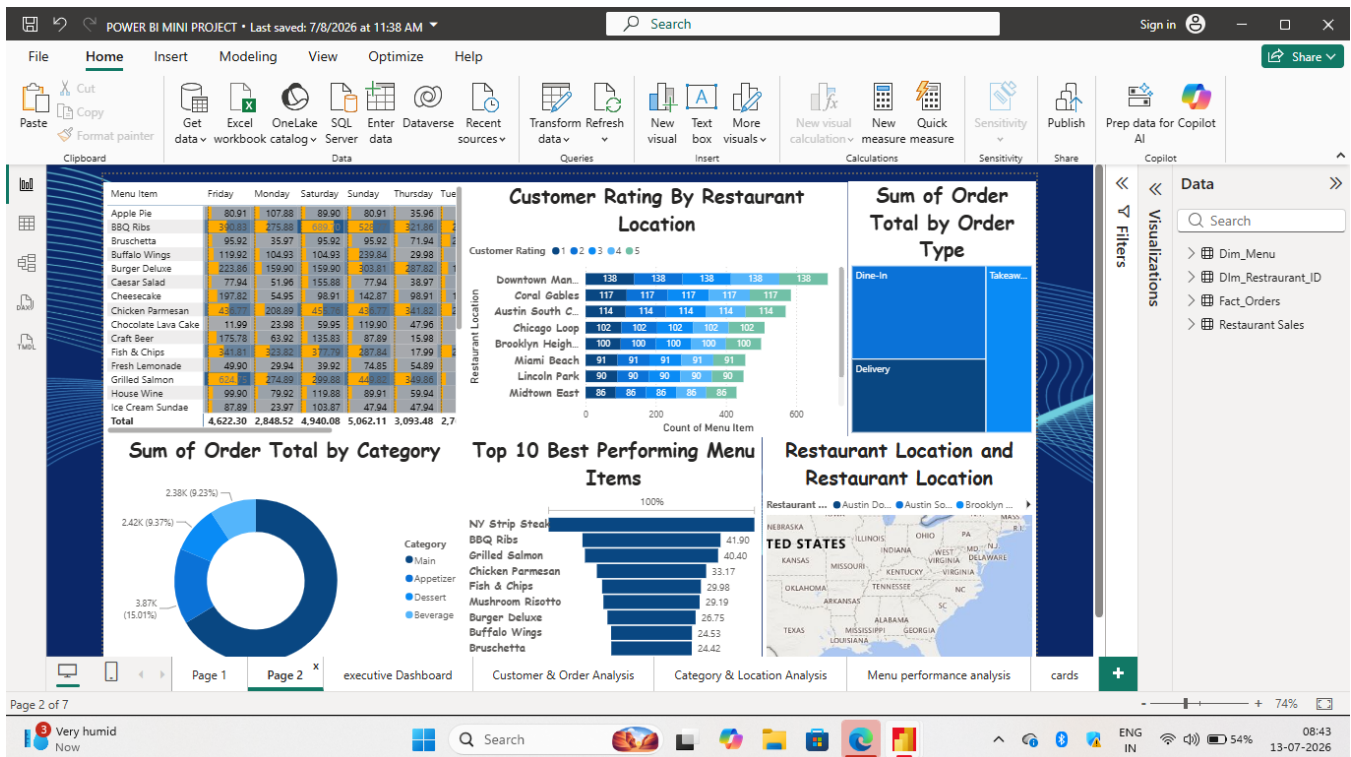
Average Rating =

AVERAGE('Restaurant Sales'[Customer Rating])

Average Order Value =

DIVIDE(
[Total Sales],
[Total Orders],
0
)





Analysis and Visualizations (Power BI)

Dashboard Features

- KPI Cards (Total Sales, Profit, Orders, Quantity)
- Year-wise Sales Analysis
- Month-wise Sales Analysis
- Day-wise Sales Analysis
- Restaurant Location-wise Sales
- Product-wise Sales Analysis
- Category-wise Sales Analysis
- Order Type Analysis
- Customer Rating Analysis
- Profit Comparison
- Top Selling Food Items

- High-Cost vs Low-Cost Food Items

Insights & Conclusions

Key Findings

- Weekend sales are significantly higher than weekday sales, indicating increased customer demand on Saturdays and Sundays.
- Monthly sales analysis reveals seasonal trends, with certain months consistently generating higher revenue than others.
- The Lincoln Park branch is performing below the overall business average and requires performance improvement initiatives.
- Dine-in orders contribute the highest percentage of total sales, making it the most preferred order type among customers.
- Best-selling menu items drive a major portion of revenue and should be prioritized for inventory management and stock availability.
- Low-selling menu items require promotional campaigns, discounts, or menu redesign to improve sales performance.
- Branch-wise performance comparison highlights differences in sales, helping identify both high-performing and underperforming locations.
- Customer rating analysis indicates that improving service quality can positively impact customer satisfaction and repeat business.
- Repeat customer analysis shows that customer loyalty plays a significant role in maintaining consistent sales growth.
- Order type analysis helps understand customer preferences, enabling better operational planning and resource allocation.
- High-demand menu items should be adequately stocked to prevent stock shortages during peak business hours.
- Weekend customer traffic requires additional staff allocation to ensure faster service and a better customer experience.
- Marketing campaigns should be targeted toward branches or locations with lower sales to improve overall business performance.
- Sales trend analysis enables accurate demand forecasting, supporting better inventory planning and workforce management.
- Monitoring customer feedback helps identify improvement areas in food quality, service efficiency, and overall customer experience.
- Combining sales, customer ratings, branch performance, and order type analysis provides valuable insights for data-driven business decisions and sustainable revenue growth.

Conclusion

This Restaurant Sales Analysis project demonstrates how Microsoft Excel and Power BI can be integrated to transform raw restaurant sales data into meaningful business insights. The interactive dashboard enables stakeholders to monitor sales performance, understand customer behavior, evaluate restaurant performance, and make informed business decisions to improve profitability and operational efficiency.